Book2 Question Bank - PDF QA

Source: Book2_Question_Bank/Essential_Histology_Question_Bank.docx

Glyph test: μ μm α β ^{3}H ± → γ Δ

This PDF is a QA artifact to verify glyph rendering, pagination, tables, and heading hierarchy.

Essential Histology: Question Bank

A Concept-Based Guide for PGME — Assessment Companion

AREMS-HY — Academic Series
Primary Reference: Junqueira 17th Edition — Official Syllabus Chapters 1,2,3,4,5,6,9,11,12,13,14,15,16,17,18,20

Essential Histology: Question Bank

A Concept-Based Guide for PGME — Assessment Companion

AREMS-HY — Academic Series
Primary Reference: Junqueira's Basic Histology: Text and Atlas — 17th Edition, by Anthony L. Mescher, McGraw Hill
Official Syllabus (Specialty Examination): Based on Junqueira 17th Chapters 1,2,3,4,5,6,9,11,12,13,14,15,16,17,18,20 — total 16 chapters. All questions based primarily on this edition and remain within official syllabus. Chapters not included not incorporated into core study plan unless specifically required.
This Question Bank is generated FROM the final textbook Essential Histology. Every question traceable to learning objective and content unit. No question tests untaught fact.
Structure per chapter: Part A Recall, Part B Understanding, Part C Application, Part D Integration, Part E High-Difficulty. Each explanation teaches why correct and why distractors wrong.

Chapter 1 — Histology & Its Methods of Study

Part A — Recall

Q1. In H&E staining, nuclei appear blue because:
A) Nucleic acids are negatively charged and bind the basic dye hematoxylin
B) Proteins bind hematoxylin
C) Eosin is a basic dye
D) DNA binds eosin
Answer: A
Explanation:** DNA/RNA phosphate groups (negative) attract positively charged hematoxylin → blue. Eosin (acidic) binds proteins (D, C reversed; B wrong).
Why distractors are wrong:
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** basophilic = nucleic acid = binds a basic dye.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to basophilic = nucleic acid = binds a basic dye.
Difficulty:** Recall

Q2. Which stain requires a frozen section?
A) Oil Red O
B) Prussian blue
C) Von Kossa
D) PAS
Answer: A
Explanation:** Lipid dyes need frozen sections because routine processing dissolves lipid. PAS, Prussian blue, Von Kossa work on paraffin.
Why distractors are wrong:
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** lipid → frozen.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to lipid → frozen.
Difficulty:** Recall

Q3. TEM exceeds light microscopy in resolution because it uses:
A) Living cells
B) Higher magnification
C) A shorter wavelength (electrons)
D) Fluorescent dyes
Answer: C

Explanation:** Resolution depends on wavelength; electrons have far shorter wavelengths than light. Magnification (B) is not the mechanism.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point:** resolution = wavelength.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to resolution = wavelength.
Difficulty:** Recall

Q4. Freeze-fracture electron microscopy specifically reveals:
A) Whole-cell 3-D shape
B) Internal organelles
C) Membrane faces (E-face/P-face) and junctional strands
D) Surface topography
Answer: C
Explanation:** Fracturing splits the bilayer's hydrophobic interior, exposing membrane faces.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point:** freeze-fracture = the membrane plane.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to freeze-fracture = the membrane plane.
Difficulty:** Recall

Q5. The Papanicolaou (Pap) stain is designed to display:
A) Nuclear chromatin detail and cytoplasmic keratinization
B) Bone mineral
C) Iron deposits
D) Lipid content
Answer: A
Explanation:** Pap reveals nuclear detail (dysplasia) and cytoplasmic maturation/keratinization — cervical cytology.
Why distractors are wrong:
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for

decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** Pap = nuclear + cytoplasmic maturation.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to Pap = nuclear + cytoplasmic maturation.
Difficulty:** Recall

Q6. The correct order of tissue processing is:
A) Fixation → dehydration → clearing → embedding → sectioning → staining
B) Clearing → fixation → embedding → sectioning
C) Fixation → staining → dehydration → embedding
D) Embedding → fixation → dehydration → staining
Answer: A
Explanation:** The canonical sequence; each step has a chemical purpose.
Why distractors are wrong:
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point:** know the pipeline order.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to know the pipeline order.
Difficulty:** Recall

Q7. Which structure stains metachromatically with toluidine blue?
A) Mast-cell granules
B) Erythrocytes
C) Elastic fibers
D) Collagen
Answer: A
Explanation:** Mast-cell heparin (sulfated GAG) polymerizes the dye → purple/red.
Why distractors are wrong:
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for

decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** mast cells = metachromatic.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to mast cells = metachromatic.
Difficulty:** Recall

Q8. A frozen section is superior to a paraffin section for:
A) Long-term storage
B) Avoiding ice artifacts
C) Best morphological detail
D) Preserving enzyme activity and lipids
Answer: D
Explanation:** Freezing bypasses the solvent/heat steps that destroy enzymes and dissolve lipids. Morphology is inferior (C, B wrong).
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** frozen = enzymes + lipids + speed.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to frozen = enzymes + lipids + speed.
Difficulty:** Recall

Q9. A researcher must count HER2 gene copies in breast cancer. The best method is:
A) PAS
B) FISH
C) Oil Red O
D) Phase-contrast
Answer: B
Explanation:** FISH counts gene copies via fluorescent probes. PAS/lipid/phase are irrelevant to DNA copy number.
Why distractors are wrong:

- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point:** gene copy number → FISH.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to gene copy number → FISH.
Difficulty:** Recall

Q10. The "clearing" agent is classically:

A) Ethanol
B) Xylene
C) Paraffin
D) Formalin
Answer: B
Explanation:** Xylene clears alcohol and makes tissue paraffin-miscible.
Why distractors are wrong:

- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point:** clearing = xylene.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to clearing = xylene.
Difficulty:** Recall

Q11. A lymph-node biopsy shows a delicate black branching mesh on a silver-stained section. The structure is:
A) Type I collagen
B) Reticular (type III) fibers
C) Elastic fibers
D) Amyloid
Answer: B

Explanation:** Reticular (type III) fibers form delicate black branching mesh on silver impregnation, seen in lymphoid stroma, liver, endocrine.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** Reticular (type III) fibers form delicate black branching mesh on silver impregnation, seen in lymph
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to Reticular (type III) fibers form delicate black branching mesh on silver impregn
Difficulty:** Recall

Q12. The resolution of a light microscope is fundamentally limited by:
A) The wavelength of visible light
B) Specimen thickness
C) Staining intensity
D) Lens quality only
Answer: A
Explanation:** Resolution ~0.2 μm limited by wavelength of visible light; electrons shorter wavelength → ~0.1 nm EM.
Why distractors are wrong:
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** Resolution ~0.2 μm limited by wavelength of visible light; electrons shorter wavelength → ~0.1 nm EM
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to Resolution ~0.2 μm limited by wavelength of visible light; electrons shorter wav
Difficulty:** Recall

Part B — Understanding

Q13. A structure that binds eosin and appears pink is best described as:

A) Basophilic
B) Acidophilic
C) Metachromatic
D) Argyrophilic

Answer: B
Explanation:** Eosinophilic/acidophilic structures (proteins, collagen) bind the acidic dye eosin → pink. Basophilic binds hematoxylin (blue); metachromatic shifts color; argyrophilic reduces silver.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** acidophilic = pink = protein.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to acidophilic = pink = protein.
Difficulty:** Understanding

Q14. Amyloid is identified by:
A) Congo red with apple-green birefringence
B) Metachromasia with toluidine blue
C) Oil Red O positivity
D) Prussian blue positivity
Answer: A
Explanation:** Congo red binds amyloid β-sheets and shows apple-green birefringence under polarized light. The others detect iron, GAGs, lipid.
Why distractors are wrong:
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** amyloid = Congo red + polarized light.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to amyloid = Congo red + polarized light.

Difficulty:** Understanding

Q15. Which technique images living, unstained cells?
A) TEM
B) Polarizing microscopy
C) H&E bright-field
D) Phase-contrast microscopy
Answer: D
Explanation:** Phase-contrast converts refractive-index differences into contrast without fixation/staining. TEM and H&E require fixed/stained specimens.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** phase-contrast = live + unstained.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to phase-contrast = live + unstained.
Difficulty:** Understanding

Q16. Prussian blue (Perls') detects:
A) Amyloid
B) Glycogen
C) Calcium
D) Iron (hemosiderin)
Answer: D
Explanation:** Perls' reaction produces blue ferric ferrocyanide from tissue iron. Calcium = Von Kossa.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** Prussian blue = iron.

Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to Prussian blue = iron.
Difficulty:** Understanding

Q17. In immunohistochemistry, the primary antibody binds:
A) A chromogen
B) The nucleus nonspecifically
C) The secondary antibody only
D) The specific antigen of interest
Answer: D
Explanation:** The primary antibody recognizes the target antigen; a labeled secondary antibody binds the primary (indirect method).
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- C) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** IHC = antibody-antigen.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to IHC = antibody-antigen.
Difficulty:** Understanding

Q18. Xylene is used in the clearing step because it:
A) Cross-links proteins
B) Removes water
C) Is miscible with both alcohol and paraffin
D) Stains nuclei
Answer: C
Explanation:** Xylene bridges alcohol (previous step) and paraffin (next step). Water removal = alcohol.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.

Key point:** clearing = the alcohol→paraffin bridge.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to clearing = the alcohol→paraffin bridge.
Difficulty:** Understanding

Q19. The most common fixative for routine light microscopy is:
A) Bouin solution
B) Glutaraldehyde
C) Formalin
D) Osmium tetroxide
Answer: C
Explanation:** Formalin (formaldehyde) is the standard LM fixative.
Glutaraldehyde/osmium = EM.
Why distractors are wrong:
- A) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- B) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
- D) is incorrect because it describes a different target/molecule or mechanism not matching the question stem. Refer to textbook Histology & Its Methods of Study section for decisive features.
Key point:** LM = formalin.
Learning Objective mapping:** Chapter 1 — Histology & Its Methods of Study — LO related to LM = formalin.
Difficulty:** Understanding

Q20. Which modality shows a bright object against a black background using oblique illumination?

Final page - End of QA sample

Glyphs verified: μ μm α β ^{3}H ± → γ
If this text shows correctly, glyphs render without tofu.